

What our Jev work can do today

This report compares the deterministic SF Pi guardrail, hosted TypeSafe Jev, and the local Gemma C11 model. It describes tested behavior and its limits.

Deterministic guardrail	Hosted Jev	Local Gemma C11
Current default	Experimental integration	Saved local candidate
Rules select risk and policy actions.	Jev selects all risk and policy actions in its mode.	Gemma selects semantic risk; code retains exact policy.
32/32 safe controls allowed in DEV175.	133/133 baseline restrictions preserved; 1/32 safe controls allowed.	105/116 model decisions correct; five unsafe allows.

Hosted Jev adds measured protection, but it is too interruptive. It caught ten authored risk cases that the deterministic engine allowed. It also asked for approval on 31 of 32 safe controls. The measured result does not support normal activation.

Gemma C11 is fast, but its safety result prevents enforcement. Its recorded warm p95 was 131 ms. It had five unsafe allows and six excess confirmations. Its recorded mode remains `off`.

The current code keeps deterministic as the default. The provider-neutral adapter requires an explicitly configured Decisions endpoint. It has no default remote address. Saved hosted results come from the earlier transport. They do not establish performance through another endpoint.

The hosted and Gemma tests use different inputs, thresholds, and policy boundaries. Their accuracy percentages do not establish a direct winner. The next useful proof is safe automatic coverage with no loss of protection.

Sources: [hosted evaluation](#), [engine decision](#), and the saved C11 report at source commit `85b12f998141081ebd9199982541a7a50520724f`.

What each engine controls

SF Pi retains blocking, approval dialogs, session grants, settings, and audit. The selected engine supplies the classification before a tool can execute.

Area	Deterministic	Hosted Jev	Gemma C11
Classification	Code rules and known tool families.	Every Pi tool call, including unfamiliar tools.	One trained semantic risk question.
Exact file policy	Code selects path eligibility, protection, and behavior.	Model interprets projected policy and local path facts.	Exact policy remains in code.
Commands and orgs	Parsed command rules, exceptions, and org-aware rules.	Model selects matches from tokens, policy, and verified facts.	Code constraints precede model risk.
Native tools	Apex, Agent Script, Data 360, SOQL, Canvas, and browser gates.	Risk, policy, disclosure, and authority questions.	Saved bridge and host hook; complete native-model workflow unproved.
Hard blocks	Code blocks cannot be bypassed by approval controls.	Any model block blocks; invalid classifications also block.	Binary model has no block choice; code retains exact blocks.
Confirmation	Explicit approval or eligible session grant. Some configured automation can approve confirmations.	Explicit human approval; legacy automation cannot bypass it.	Helper returns allow, confirm, or abstain; host owns enforcement.
Audit	Gated decisions; unmatched calls can pass without an entry.	Each classified outcome, with bounded model evidence.	Saved scoring and controlled host proofs; no normal-profile activation.

The deterministic engine covers much more than shell strings. Its known native gates handle execution, durable writes, disclosure, and committing browser gestures. Unmatched calls can pass through. It is not a general filesystem or shell sandbox.

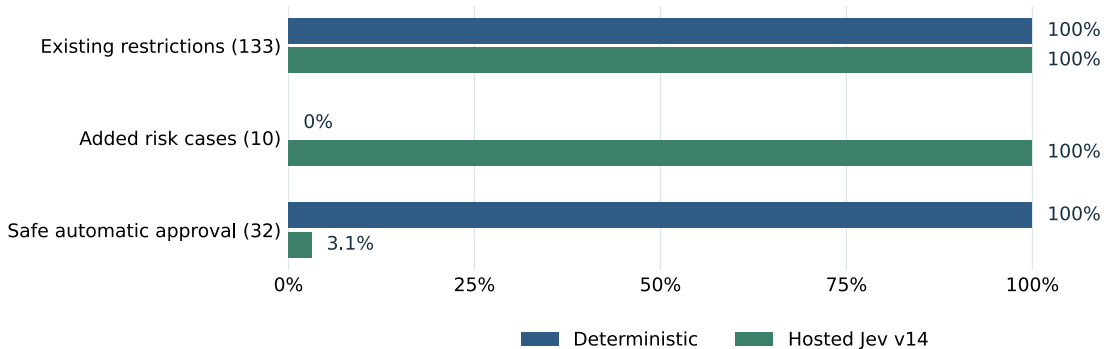
Hosted Jev has sole policy authority in its selected mode. There is no deterministic safety floor or automatic fallback. Exact configured patterns therefore lose their code matching guarantee, even when a finite test passes.

Sources: [Safety Kernel](#), [native registry](#), [Jev adapter](#), and [extension contract](#).

Protection and useful approval

The same 175 development inputs supplied this comparison. Authored expected actions were 32 allows, 132 confirmations, and eleven blocks. Some labels need review. These are consumed development inputs, not a fresh independent test.

Measure	Deterministic	Hosted Jev v14
Exact action against authored labels	165/175 (94.3%)	144/175 (82.3%)
Existing restrictions preserved	Reference: 133	133/133 (100%)
Existing hard blocks preserved	Reference: 11	11/11 (100%)
Added authored risks restricted	0/10	10/10 (100%)
Safe automatic recommendations	32/32 (100%)	1/32 (3.1%)
Safe controls interrupted	0/32	31/32 (96.9%)
Allow / confirm / block	42 / 122 / 11	1 / 163 / 11



Each counted Jev catch had an actual non-allow model choice. All eleven hard blocks had actual block choices. Failed calls and low allow probabilities alone did not count as recognition. All 175 v14 responses passed validation.

The added cases cover wrapped or nested production deployment, production REST writes, shell file access, external deletion, force-with-lease, broad SOQL disclosure, and unfamiliar external writes. The fixed labels are test judgments. They do not prove that every described effect occurred.

The runtime uses the smaller v11 request representation. Its saved test had the same action counts, protection counts, and 1/32 safe automatic result. The larger v14 command question remains a historical experiment.

Source: full evaluation and label limits.

How the current Jev path works

1. Capture the original call identity and configured transport identity locally.
2. Build bounded operation metadata and resolve required local facts.
3. Ask applicable independent Choice questions in one Decisions request.
4. Validate the exact answer set, identity, probabilities, size, and deadline.
5. Block, allow, or ask for explicit approval; record the outcome before release.

Question	What Jev judges
Risk	Executable effects and essential uncertainty; always included.
File policy	Path eligibility, rule precedence, exemptions, and access.
Command policy	Exact token matches, ordered restrictions, and allow exceptions.
Org policy	Command structure, rule order, org facts, and local exclusions.
Disclosure	Credential or data output and transfer effects.
Authority	Browser target, focus, role, and committing gestures.

Any block answer blocks execution. Automatic approval requires complete context and every answer choosing allow with $P(\text{allow}) \geq 0.99$. Other valid results require human approval. Failures block. There are no retries.

The transport requests `typesafe/jev-1.13`. It requires resolved model `typesafe/jev-1.13-20260917` and provider `TypeSafe`. A configured endpoint must implement this Decisions contract. An ordinary chat API is not interchangeable.

Configure `SF_GUARDRAIL_JEV_ENDPOINT` and either `SF_GUARDRAIL_JEV_API_KEY` or `SF_GUARDRAIL_JEV_API_KEY_FILE`. A documentation placeholder is `https://decisions.example.invalid/v1/decisions`. Replace it with a compatible endpoint before use. Missing or invalid endpoint configuration blocks.

The local transport hash binds grants to the canonical endpoint and model pins. Execution checks reject a changed endpoint. Audit and UI omit its value. Raw commands, code, scripts, query bodies, file bodies, credentials, transcripts, Canvas contents, and full browser pages remain local. Token IDs reveal equality and policy membership; they are not cryptographic secrecy.

Sources: [client](#), [request and decision adapter](#), and [TypeSafe Choice contract](#).

What your trained model provides

The candidate is `jev/c11-step-256`, based on `google/gemma-3-1b-it`. It is a fully merged F16 GGUF with a pinned local native scorer. The model file remains present at 2,006,573,408 bytes. This report checked its metadata and prior verification receipt. It did not rehash the weights or rerun inference.

Training used a new LoRA adapter on query and value projections in all 26 layers, with rank 16. The base weights stayed frozen. The FIT campaign used 327 rows and 86 safe/risky pairs. Its loss combined classification, margins, and pair separation. The selected export is step 256. The campaign used no VALID or TEST rows and called no teacher model in the inspected training path.

The scorer reads selected next-token label logits for allow and confirm. It generates no explanation. Its helper also returns abstain for insufficient allow score. The cutoff is `0.955913273071778`, an uncalibrated threshold.

Recorded C11 result	Value
Correct model decisions	105/116 (90.52%)
Correct decisions including code	149/160 (93.13%)
Model-eligible safe controls allowed	69/75 (92.0%)
Model-eligible risky cases allowed	5/41 (12.2%)
Unsafe allows / excess confirmations	5 / 6
Warm p95 / maximum	131.17 / 145.56 ms
Cold initialization	1,364.09 ms
Scoring deadline / warm p95 target	750 / 500 ms
Independent TEST score	Absent
Qualification / enforcement / mode	False / false / off

The overall score includes 44 correct non-model decisions. It does not measure model accuracy on all 160 inputs. The five unsafe categories include remote scripts, encoded execution, environment disclosure, external HTTP writes, and file-image overwrite. These are scored decisions; the operations did not run.

An archived SF Pi hook retained code policy before local model risk. Native replay used the actual model. Separate SDK proofs used scripted predictions. The native-model Pi workflow test was skipped. A complete live workflow remains unproved. The portable demo provides advisory scoring, not enforcement.

Sources: saved C11 stop report, step-256 diagnostic records, model card, portable registry, and archived host commit `a4ba5fe5f86bc0cb01ab85c037bb05114a26ff8a`.

Readiness depends on more than accuracy

The hosted v11 test reported remote p50 of 407 ms and p95 of 607 ms. V14 reported 476 ms and 612 ms. Both p95 values exceed the initial 500 ms target. The waits exclude real fact preparation and human approval. The normal total classification deadline remains 1,500 ms. The tests used ten seconds.

Reported response cost was \$0.0384 for v11 and \$0.0411 for v14. Known reported task response cost through v14 was \$0.3408. Failed-call billing remains unknown. This is not a complete account bill. Local C11 has no provider fee per call; hardware, setup, memory, and energy costs were not measured here.

The earlier v11 source integration reproduced all 175 saved request bodies exactly, offline. Its 73 focused risk tests passed. Its full CI passed 4,946 tests and skipped 39. Its lint passed.

The current provider-neutral source passed 260 focused tests. Full CI passed 4,992 tests and skipped 39. Lint passed. Its current offline proof matched all 175 v11 request bodies with zero external calls. These checks establish source compatibility and request fidelity. No new live endpoint proof exists. The checks do not establish a new endpoint's performance or model quality.

One real hosted SDK smoke released an inert counter tool after an audited allow. Two other smokes blocked on deadline. No Salesforce or browser mutation was established. The new generic transport requires its own live proof.

The next steps are:

1. Keep every observed baseline restriction and hard block while reducing false confirmations.
2. Review safe labels for legacy CLI flags, deployment jobs, local tracking, OAuth refresh, and withheld query contents.
3. Measure model recognition separately from confirmation caused only by a low probability.
4. Run a new frozen test after independent review; retain failures in every denominator.
5. Measure the full hook with real fact resolution before normal activation.

The council selected a small neutral-option-name test: 26 cases, four arms, and at most 104 calls. Its 21 scripted checks passed. It made no live calls. Neutral names are a test idea, not a proven improvement. The 26/32 utility gate was proposed by an agent; the user has not approved it as qualification.

TypeSafe documents sensitivity to indirection, numeric representations, contradictory guidance, and irrelevant context. It recommends code for work that code can compute. Sole Jev policy authority remains the chosen boundary. [Model limits](#). Probability shape is not domain calibration. [Confidence guidance](#).

Sources and handoff

The product work belongs to the existing `sf-pi` repository and its existing `sf-guardrail` extension. It is delivered on a feature branch. It is not a separate extension repository. Source defaults remain deterministic.

The following public files supply implementation and result detail:

- `Hosted evaluation`: all attempts, old failures, labels, latency, cost, and activation limits.
- `Council report`: expert positions, disagreement, and the bounded next test.
- `ADR 0118`: engine boundary and approval contract.
- `Client and risk adapter`: transport, validation, questions, and decisions.
- `Safety Kernel` and `native registry`: deterministic behavior.
- `TypeSafe Choice`, `model limits`, and `confidence`: external model guidance.

The C11 facts were checked against retained source commit `85b12f998141081ebd9199982541a7a50520724f`, its saved stop report and diagnostic records, and the current portable model card. Those saved records are separate from the hosted DEV175 set. No sealed C11 TEST corpus was opened.

The current protocol is version 8. The current offline proof matched all 175 request bodies with the saved v11 bodies. It made zero external calls.

Frozen evidence	SHA-256
Hosted v11 receipt	eb116b58bffc2e8b96695b9fadfa2fe87a5cab834171b27b10c0a33c56e0f00
Hosted v14 receipt	7cd6a4f50048004f18d3d371e7bd2cf9f73de7437a1d139f070a26cd14948e5e
Earlier v11 source proof	25f6a3e4f1a78d3fdaa0b79fee9496c90f973efbd68dc2863d6bf24393da123f
Current protocol, version 8	42c2049ed4ef29a5e6baa442dee3fd68d77b3ff163901782a998d66b8fc214fd
Current offline v11 byte proof	b43a3005f7b191d1abad97642294b00833b31aa573624b8e62dde1edc9440e3f
C11 merged model, prior verified identity	8930f0028412060fe5f39d590fe5906ca2ed86c5a6613473512580d62994a053

Raw hosted receipts and test-only probes remain local evidence. This report publishes aggregate results and generic examples. It contains no credential, private endpoint, raw tool payload, or private repository URL.

Use the deterministic engine for the current default. Hosted Jev supplies an exercised experimental path with broader observed risk detection. Gemma C11 supplies a fast, portable local classifier with known unsafe decisions. Neither model result qualifies a replacement for normal enforcement today.